

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

```
adithya@LAPTOP-07A78KDI:~/OS$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu InRelease
Get:2 http://archive.ubuntu.com/ubuntu InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu InRelease
Get:5 http://security.ubuntu.com/ubuntu InRelease [383 kB]
Get:6 http://archive.ubuntu.com/ubuntu InRelease [458 kB]
Get:7 http://security.ubuntu.com/ubuntu InRelease [92.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu InRelease [112 kB]
Get:9 http://security.ubuntu.com/ubuntu InRelease [46.6 kB]
Get:10 http://archive.ubuntu.com/ubuntu InRelease [97.8 kB]
Get:11 http://security.ubuntu.com/ubuntu InRelease [154 kB]
Get:12 http://archive.ubuntu.com/ubuntu InRelease [237 kB]
Get:13 http://archive.ubuntu.com/ubuntu InRelease [77.8 kB]
Get:14 http://security.ubuntu.com/ubuntu InRelease [50.0 kB]
]
Get:15 http://archive.ubuntu.com/ubuntu InRelease [182 kB]
Get:16 http://security.ubuntu.com/ubuntu InRelease [43.1 kB]
Get:17 http://archive.ubuntu.com/ubuntu InRelease [341 kB]
Get:18 http://security.ubuntu.com/ubuntu InRelease [324 kB]
Get:19 http://archive.ubuntu.com/ubuntu InRelease [65.7 kB]
]
Get:20 http://security.ubuntu.com/ubuntu InRelease [61.6 kB]
Get:21 http://archive.ubuntu.com/ubuntu InRelease [11.3 kB]
]
Get:22 http://archive.ubuntu.com/ubuntu InRelease [3032 B]
Fetched 3016 kB in 7s (407 kB/s)
20 packages can be upgraded. Run 'apt list --upgradable' to see them.
adithya@LAPTOP-07A78KDI:~/OS$
```

```
adithya@LAPTOP-07A78KDI:~/OS$ sudo apt upgrade
Upgrading:
  alsa-ucm-conf      libjavascriptcoregtk-6.0-1  libudev1      systemd-cryptsetup
  apparmor           libnss-systemd              libwebkitgtk-6.0-4  systemd-resolved
  cloud-init         libpam-systemd              libyelp-1-0        systemd-sysv
  cloud-init-base    libsystemd-shared           linux-libc-dev     udev
  libapparmor1       libsystemd0                 systemd          yelp

Summary:
  Upgrading: 20, Installing: 0, Removing: 0, Not Upgrading: 0
  13 standard LTS security updates
  Download size: 45.2 MB
  Space needed: 42.0 kB / 1023 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu InRelease [481 kB]
Get:2 http://archive.ubuntu.com/ubuntu InRelease [9250 B]
Get:3 http://archive.ubuntu.com/ubuntu InRelease [299 kB]
Get:4 http://archive.ubuntu.com/ubuntu InRelease [274 kB]
Get:5 http://archive.ubuntu.com/ubuntu InRelease [2462 kB]
Get:6 http://archive.ubuntu.com/ubuntu InRelease [3129 kB]
11% [6 systemd 15.6 kB/3129 kB 0%] 576 kB/s 1min 7s^Get:7 http://archive.ubuntu.com/ubuntu InRelease [124 kB]
18% [7 systemd-cryptsetup 14.2 kB/124 kB 11%] 576 kB/s 1min 6s^Get:8 http://archive.ubuntu.com/ubuntu InRelease [1501 kB]
20% [8 udev 308 kB/1501 kB 21%] 576 kB/s 1min 6s^Get:9 http://archive.ubuntu.com/ubuntu InRelease [151 kB]
24% [Working] 576 kB/s 1min 3s^Get:10 http://archive.ubuntu.com/ubuntu InRelease [51.3 kB]
26% [Working] 576 kB/s 1min 3s^Get:11 http://archive.ubuntu.com/ubuntu InRelease [227 kB]
26% [12 apparmor 1156 B/380 kB 0%] 576 kB/s 1min 3s^Get:12 http://archive.ubuntu.com/ubuntu InRelease [106 kB]
Get:13 http://archive.ubuntu.com/ubuntu InRelease [587 kB]
Get:14 http://archive.ubuntu.com/ubuntu InRelease [2110 B]
```

```
adithya@LAPTOP-07A78KDI:~/OS$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
adithya@LAPTOP-07A78KDI:~/OS$ nano hello.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc hello.c -o hello
adithya@LAPTOP-07A78KDI:~/OS$ ./hello
Parent Process Started
Parent PID : 2769
Parent PPID : 329

Parent created Child
Child PID : 2770
Child Process Started
Child PID : 2770
Child PPID : 2769

Child is executing 'ls -l' using exec()

total 264
drwxr-xr-x 6 adithya adithya 4096 Aug 1 08:32 GIT
drwxr-xr-x 2 adithya adithya 4096 Aug 5 07:09 OSSP
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:54 Skill1
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill10
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill11
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill12
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill2
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill3
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill4
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill5
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill6
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill7
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill8
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill9
-rwxr-xr-x 1 adithya adithya 16552 Aug 8 09:21 client
-rw-r--r-- 1 adithya adithya 1348 Aug 8 09:21 client.c
-rwxr-xr-x 1 adithya adithya 16056 Jul 28 03:43 file
-rw-r--r-- 1 adithya adithya 226 Jul 28 03:43 file.c
drwxr-xr-x 2 adithya adithya 4096 Jul 31 15:48 fork
-rwxr-xr-x 1 adithya adithya 16288 Aug 14 08:59 hello
-rw-r--r-- 1 adithya adithya 986 Aug 14 08:59 hello.c
-rwxr-xr-x 1 adithya adithya 16264 Aug 8 08:37 practical2
-rw-r--r-- 1 adithya adithya 1420 Aug 8 08:52 practical2.c
-rw-r--r-- 1 adithya adithya 1218 Aug 8 08:48 practical3.c
-rwxr-xr-x 1 adithya adithya 16344 Aug 8 08:56 practical4
-rw-r--r-- 1 adithya adithya 1787 Aug 8 08:59 practical4.c
-rwxr-xr-x 1 adithya adithya 16432 Aug 8 09:15 practical5
-rw-r--r-- 1 adithya adithya 1548 Aug 8 09:13 practical5.c
-rwxr-xr-x 1 adithya adithya 16136 Aug 8 08:54 process_states
-rw-r--r-- 1 adithya adithya 413 Aug 8 08:54 process_states.c
-rw-r--r-- 1 adithya adithya 24 Aug 8 08:40 sample.txt
-rwxr-xr-x 1 adithya adithya 16536 Aug 8 09:21 server
-rw-r--r-- 1 adithya adithya 1728 Aug 8 09:20 server.c
-rw-r--r-- 1 adithya adithya 40 Aug 8 08:38 source.txt
-rwxr-xr-x 1 adithya adithya 16208 Aug 8 09:05 zombie
-rw-r--r-- 1 adithya adithya 698 Aug 8 09:05 zombie.c

Child process completed.
Parent process completed.
adithya@LAPTOP-07A78KDI:~/OS$ |
```

```
adithya@LAPTOP-07A78KDI:~/OS$ mkdir os-lab
cd os-lab
```

```
mkdir src include bin
touch README.md Makefile
```

```
ls -R
.:
Makefile README.md bin include src
```

```
./bin:
```

```
./include:
```

```
./src:
```

```
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ |
```

```
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/adithya/OS/os-lab/.git/
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ git status
On branch master
```

```
No commits yet
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  Makefile
  README.md
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ git add .
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ git commit -m "Initial project structure"
Author identity unknown
```

```
*** Please tell me who you are.
```

```
Run
```

```
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
```

```
to set your account's default identity.
Omit --global to set the identity only in this repository.
```

```
fatal: empty ident name (for <adithya@LAPTOP-07A78KDI.localdomain>) not allowed
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ |
```

```
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
adithya      329     328  0 08:53 pts/0    00:00:00 -ba
adithya     2892     329  0 09:02 pts/0    00:00:00 ps
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ |
```

```

adithya@LAPTOP-07A78KDI:~/OS/os-lab$ ps -f
UID          PID     PPID  C  STIME TTY          TIME CMD
adithya      329      328  0   08:53 pts/0        00:00:00 -bash
adithya     2892      329  0   09:02 pts/0        00:00:00 ps -f
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ nano hello.c
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ nano Makefile
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ make
make: *** No rule to make target 'src/main.c', needed by 'bin/main'.  Stop.
adithya@LAPTOP-07A78KDI:~/OS/os-lab$ make clean
rm -f bin/main
adithya@LAPTOP-07A78KDI:~/OS/os-lab$

```

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("=== Process Demonstration ===\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n\n", getppid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("=== Child Process ===\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("\nChild executing 'ls -l' using exec()\n\n");

        execlp("ls", "ls", "-l", NULL);

        // Executes only if exec() fails
        perror("exec failed");
        exit(1);
    }

    else {
        // Parent process
        printf("=== Parent Process ===\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("\nParent waiting for child...\n");

        wait(NULL);

        printf("Child completed.\n");
        printf("Parent completed.\n");
    }

    return 0;
}

```

```

adithya@LAPTOP-07A78KDI:~/OS$ nano process_demo.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc process_demo.c -o process_demo
adithya@LAPTOP-07A78KDI:~/OS$ ./process_demo
=== Process Demonstration ===
Parent PID : 3015
Parent PPID : 329

=== Parent Process ===
Parent PID : 3015
Child PID : 3016

Parent waiting for child...
=== Child Process ===
Child PID : 3016
Child PPID : 3015

Child executing 'ls -l' using exec()

total 288
drwxr-xr-x 6 adithya adithya 4096 Aug 1 08:32 GIT
drwxr-xr-x 2 adithya adithya 4096 Aug 5 07:09 OSSP
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:54 Skill11
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill110
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill111
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill112
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill12
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill13
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill14
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill15
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill16
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill17
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill18
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill19
-rwxr-xr-x 1 adithya adithya 16552 Aug 8 09:21 client
-rw-r--r-- 1 adithya adithya 1348 Aug 8 09:21 client.c
-rwxr-xr-x 1 adithya adithya 16056 Jul 28 03:43 file
-rw-r--r-- 1 adithya adithya 226 Jul 28 03:43 file.c
drwxr-xr-x 2 adithya adithya 4096 Jul 31 15:48 fork
-rwxr-xr-x 1 adithya adithya 16288 Aug 14 08:59 hello
-rw-r--r-- 1 adithya adithya 986 Aug 14 08:59 hello.c
drwxr-xr-x 6 adithya adithya 4096 Aug 14 09:03 os-lab
-rwxr-xr-x 1 adithya adithya 16264 Aug 8 08:37 practical2
-rw-r--r-- 1 adithya adithya 1420 Aug 8 08:52 practical2.c
-rw-r--r-- 1 adithya adithya 1218 Aug 8 08:48 practical3.c
-rwxr-xr-x 1 adithya adithya 16344 Aug 8 08:56 practical4
-rw-r--r-- 1 adithya adithya 1787 Aug 8 08:59 practical4.c
-rwxr-xr-x 1 adithya adithya 16432 Aug 8 09:15 practical5
-rw-r--r-- 1 adithya adithya 1548 Aug 8 09:13 practical5.c
-rwxr-xr-x 1 adithya adithya 16296 Aug 14 09:05 process_demo
-rw-r--r-- 1 adithya adithya 1072 Aug 14 09:05 process_demo.c
-rwxr-xr-x 1 adithya adithya 16136 Aug 8 08:54 process_states
-rw-r--r-- 1 adithya adithya 413 Aug 8 08:54 process_states.c
-rw-r--r-- 1 adithya adithya 24 Aug 8 08:40 sample.txt
-rwxr-xr-x 1 adithya adithya 16536 Aug 8 09:21 server
-rw-r--r-- 1 adithya adithya 1728 Aug 8 09:20 server.c
-rw-r--r-- 1 adithya adithya 40 Aug 8 08:38 source.txt
-rwxr-xr-x 1 adithya adithya 16208 Aug 8 09:05 zombie
-rw-r--r-- 1 adithya adithya 698 Aug 8 09:05 zombie.c
Child completed.
Parent completed.

```

```

adithya@LAPTOP-07A78KDI:~/OS$ pid = fork();
-bash: syntax error near unexpected token '('
adithya@LAPTOP-07A78KDI:~/OS$ |

```

```

adithya@LAPTOP-07A78KDI:~/OS$ pid = fork();
-bash: syntax error near unexpected token '('
adithya@LAPTOP-07A78KDI:~/OS$ getpid()
> getppid()
-bash: syntax error near unexpected token 'getppid'
adithya@LAPTOP-07A78KDI:~/OS$ |

```

```

adithya@LAPTOP-07A78KDI:~/OS$ getppid()
> getppid()
-bash: syntax error near unexpected token `getppid'
adithya@LAPTOP-07A78KDI:~/OS$ execlp("ls", "ls", "-l", NULL);
-bash: syntax error near unexpected token `ls', '
adithya@LAPTOP-07A78KDI:~/OS$ pstree -p
systemd(1)
├─agetty(214)
├─chronyd-starter(106)─chronyd(217)─chronyd(232)
├─cron(107)
├─dbus-daemon(108)
├─init-systemd(Ub(2))
│   ├──SessionLeader(327)─Relay(329)(328)─bash(329)─pstree(3079)
│   │   ├──init(6)─{init}(7)
│   │   ├──login(330)─bash(405)
│   │   └──{init-systemd(Ub)}(8)
├─networkd-dispat(123)
├─polkitd(601)
│   ├──{polkitd}(603)
│   ├──{polkitd}(604)
│   └──{polkitd}(605)
├─rsyslogd(2616)
│   ├──{rsyslogd}(2617)
│   ├──{rsyslogd}(2618)
│   └──{rsyslogd}(2619)
├─systemd(373)─{sd-pam}(375)
├─systemd-journal(1929)
├─systemd-journal(160)
├─systemd-resolve(2505)
├─systemd-udevd(2433)
└─unattended-upgr(241)─{unattended-upgr}(288)

adithya@LAPTOP-07A78KDI:~/OS$ ps
  PID TTY          TIME CMD
   329 pts/0    00:00:00 bash
  3089 pts/0    00:00:00 ps

adithya@LAPTOP-07A78KDI:~/OS$ sudo apt update
sudo apt install strace
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
All packages are up to date.
strace is already the newest version (6.19+ds-0ubuntu5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
adithya@LAPTOP-07A78KDI:~/OS$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mpers
adithya@LAPTOP-07A78KDI:~/OS$

```

```

adithya@LAPTOP-07A78KDI:~/OS$ strace -e trace-process ./process_demo
execve("./process_demo", ["/process_demo"], 0x7ff6b2f17140 /* 29 vars */) = 0
=== Process Demonstration ===
Parent PID : 3268
Parent PPID : 3266

clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x78cc69e22a10) = 3269
=== Child Process ===
Child PID : 3269
Child PPID : 3268

Child executing 'ls -l' using exec()

=== Parent Process ===
Parent PID : 3268
Child PID : 3269

Parent waiting for child...
wait4(-1total 288
drwxr-xr-x 6 adithya adithya 4096 Aug 1 08:32 GIT
drwxr-xr-x 2 adithya adithya 4096 Aug 5 07:09 OSSP
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:54 Skill11
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill110
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill111
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill112
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill12
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill13
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill14
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill5
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill6
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill7
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill8
drwxr-xr-x 2 adithya adithya 4096 Aug 3 05:31 Skill9
-rwxr-xr-x 1 adithya adithya 16552 Aug 8 09:21 client.c
-rw-r--r-- 1 adithya adithya 1348 Aug 8 09:21 client.c
-rwxr-xr-x 1 adithya adithya 16956 Jul 28 03:43 file
-rw-r--r-- 1 adithya adithya 226 Jul 28 03:43 file.c
drwxr-xr-x 2 adithya adithya 4096 Jul 31 15:08 fork
-rwxr-xr-x 1 adithya adithya 16288 Aug 14 08:59 hello
-rw-r--r-- 1 adithya adithya 986 Aug 14 08:59 hello.c
drwxr-xr-x 6 adithya adithya 4096 Aug 14 09:03 os-lab
-rwxr-xr-x 1 adithya adithya 16204 Aug 8 08:37 practical2
-rw-r--r-- 1 adithya adithya 1420 Aug 8 08:02 practical2.c
-rw-r--r-- 1 adithya adithya 1218 Aug 8 08:48 practical3.c
-rwxr-xr-x 1 adithya adithya 16344 Aug 8 08:56 practical4
-rw-r--r-- 1 adithya adithya 1787 Aug 8 08:59 practical4.c
-rwxr-xr-x 1 adithya adithya 16432 Aug 8 09:15 practical5
-rw-r--r-- 1 adithya adithya 1548 Aug 8 09:13 practical5.c
-rwxr-xr-x 1 adithya adithya 16296 Aug 14 09:05 process_demo
-rw-r--r-- 1 adithya adithya 1972 Aug 14 09:07 process_demo.c
-rwxr-xr-x 1 adithya adithya 16136 Aug 8 08:54 process_states
-rw-r--r-- 1 adithya adithya 413 Aug 8 08:54 process_states.c
-rw-r--r-- 1 adithya adithya 24 Aug 8 08:40 sample.txt
-rwxr-xr-x 1 adithya adithya 16536 Aug 8 09:21 server
-rw-r--r-- 1 adithya adithya 1728 Aug 8 09:20 server.c
-rw-r--r-- 1 adithya adithya 40 Aug 8 08:38 source.txt
-rwxr-xr-x 1 adithya adithya 16208 Aug 8 09:05 zombie
-rw-r--r-- 1 adithya adithya 698 Aug 8 09:05 zombie.c
, NULL, 0, NULL)
= 3269
--- SIGCHLD (si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=3269, si_uid=1000, si_status=0, si_etime=0, si_stime=0) ---
Child completed.
Parent completed.
exit_group(0)
= 7
+++ exited with 0 +++
adithya@LAPTOP-07A78KDI:~/OS$

```